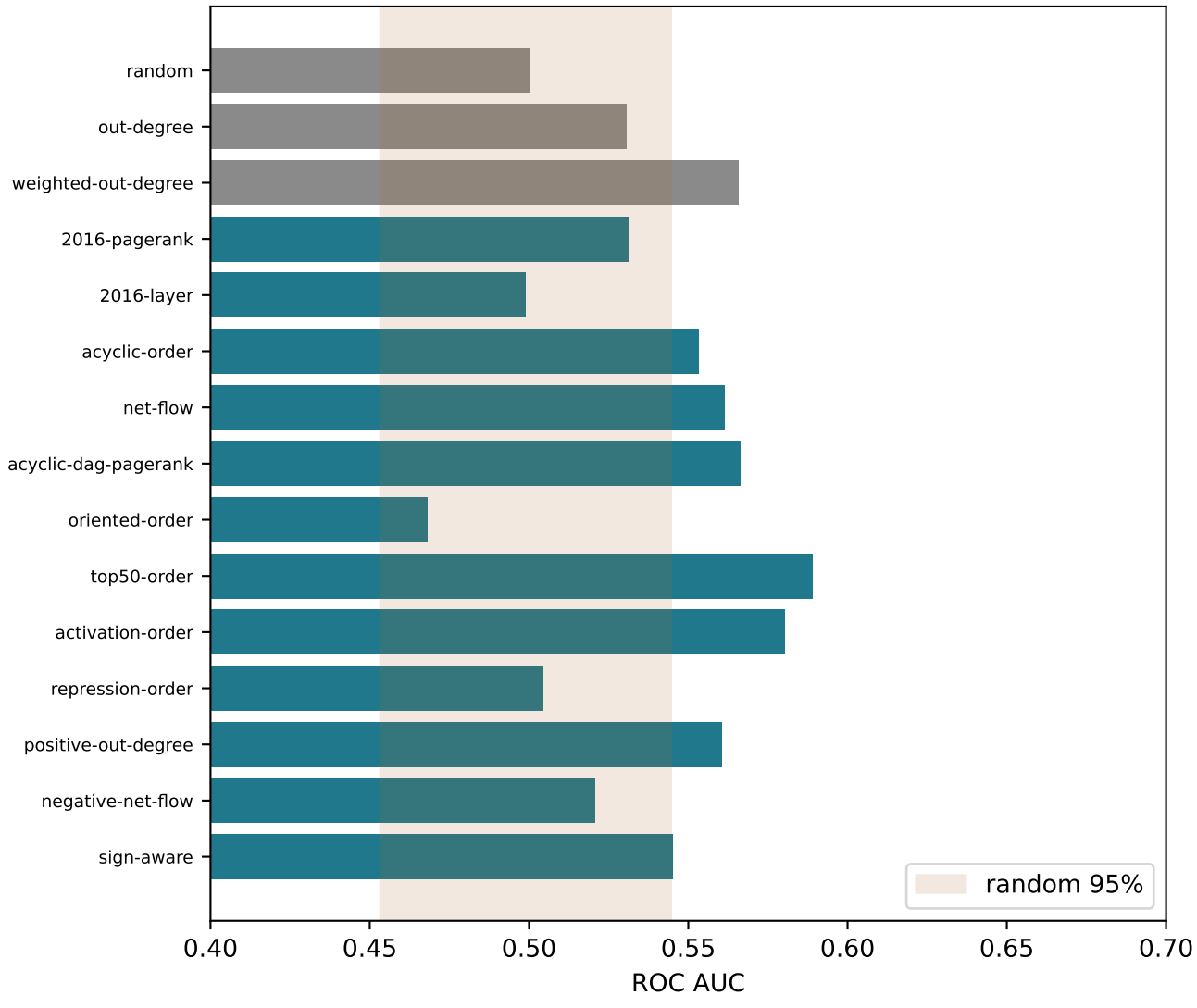
Regulator ranking against reference 'regulators'

2026-11-07 · 740 genes ranked, 200 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	27.0 (19–35)	0.500 (0.453–0.545)				
out-degree	27	0.531	0.548	0.098	-0.035 (-0.044, -0.028)	< 0.001
weighted-out-degree	35	0.566	0.034	0.002	+0.000 (+0.000, +0.000)	
2016-pagerank	33	0.531	0.084	0.093	-0.035 (-0.069, +0.000)	0.056
2016-layer	65	0.499	< 0.001	0.518	-0.067 (-0.104, -0.029)	< 0.001
acyclic-order	34	0.553	0.052	0.009	-0.013 (-0.036, +0.011)	0.304
net-flow	35	0.561	0.034	0.003	-0.005 (-0.014, +0.005)	0.352
acyclic-dag-pagerank	35	0.566	0.034	0.002	+0.000 (-0.016, +0.017)	0.948
oriented-order	17	0.468	0.995	0.911	-0.098 (-0.128, -0.066)	< 0.001
top50-order	44	0.589	< 0.001	< 0.001	+0.023 (-0.015, +0.063)	0.244
activation-order	46	0.580	< 0.001	< 0.001	+0.014 (-0.014, +0.042)	0.298
repression-order	29	0.504	0.357	0.438	-0.062 (-0.094, -0.027)	0.002
positive-out-degree †	40	0.561	0.002	0.005	-0.005 (-0.022, +0.012)	0.566
negative-net-flow †	25	0.521	0.725	0.200	-0.045 (-0.065, -0.026)	< 0.001
sign-aware †	29	0.545	0.357	0.022	-0.021 (-0.032, -0.011)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

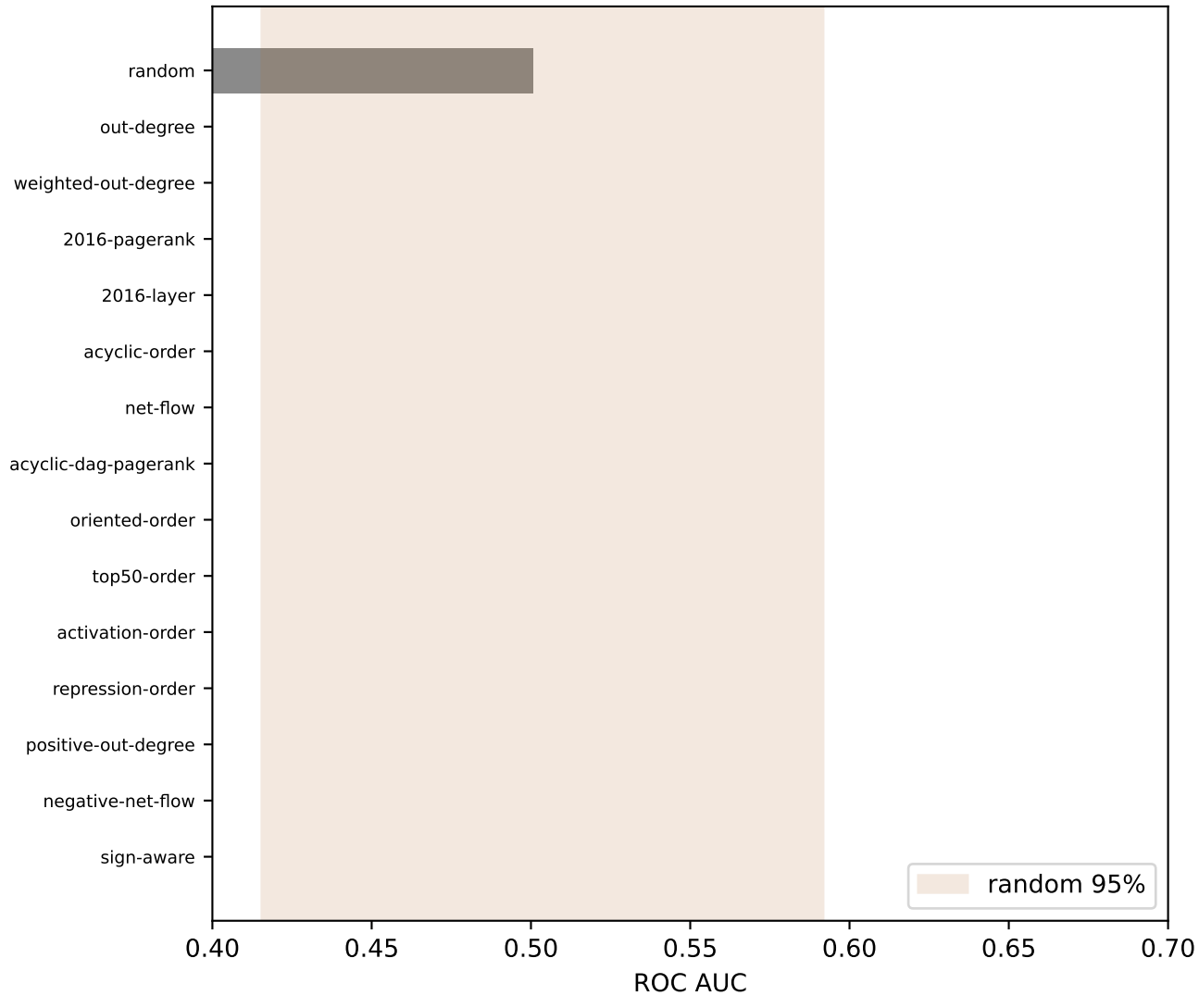


Regulator ranking against reference 'top-layer'

2026-11-07 · 740 genes ranked, 42 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	5.7 (2–10)	0.501 (0.416–0.592)				
out-degree	3	0.279	0.933	1.000	-0.031 (-0.043, -0.020)	< 0.001
weighted-out-degree	3	0.311	0.933	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	1	0.327	0.997	1.000	+0.016 (-0.049, +0.091)	0.644
2016-layer	6	0.321	0.520	1.000	+0.010 (-0.056, +0.074)	0.728
acyclic-order	2	0.262	0.980	1.000	-0.049 (-0.086, -0.012)	0.010
net-flow	3	0.306	0.933	1.000	-0.004 (-0.024, +0.017)	0.684
acyclic-dag-pagerank	2	0.281	0.980	1.000	-0.030 (-0.048, -0.012)	< 0.001
oriented-order	0	0.215	1.000	1.000	-0.096 (-0.144, -0.049)	< 0.001
top50-order	1	0.299	0.997	1.000	-0.012 (-0.057, +0.033)	0.620
activation-order	3	0.288	0.933	1.000	-0.022 (-0.060, +0.015)	0.252
repression-order	1	0.308	0.997	1.000	-0.003 (-0.068, +0.070)	0.946
positive-out-degree †	3	0.311	0.933	1.000	+0.001 (-0.020, +0.020)	0.974
negative-net-flow †	2	0.290	0.980	1.000	-0.021 (-0.053, +0.015)	0.240
sign-aware †	3	0.293	0.933	1.000	-0.018 (-0.039, +0.003)	0.100

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	3,796	0.738 (22.97×)	27.49×	16.66×	1.00 (0.58)
resolve_reciprocals → drop_sinks → prune_triangles	3,758	0.089 (17.41×)	26.04×	14.64×	1.00 (0.54)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	3,118	0.031 (10.22×)	14.00×	9.34×	1.00 (0.61)
absolute	3,796	0.738 (22.97×)			
absolute → minimum_feedback_order	3,796	0.651 (22.12×)			
orient_reciprocals ratio=2.0 → absolute	3,796	0.479 (19.12×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	3,796	0.479 (19.12×)			
top_outgoing k=50 → absolute	3,796	0.640 (32.38×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,796	0.563 (31.75×)			
keep_sign sign=1 → absolute	3,796	0.515 (27.49×)			
keep_sign sign=1 → absolute → minimum_feedback_order	3,796	0.461 (26.86×)			
keep_sign sign=-1 → absolute	3,796	0.224 (16.66×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	3,796	0.187 (15.41×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 45,475 edges (20,452 negative), largest strong component 714.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 45,475 edges (20,452 negative), largest strong component 714.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,229 edges (5,462 negative), largest strong component 139.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,229 edges (5,462 negative), largest strong component 139; DAG keeps 3,622 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 45,475 edges (0 negative), largest strong component 714; DAG keeps 40,165 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 45,475 edges (0 negative), largest strong component 714.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 45,475 edges (0 negative), largest strong component 714; DAG keeps 40,165 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 35,309 edges (0 negative), largest strong component 1; DAG keeps 35,309 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 28,693 edges (0 negative), largest strong component 714; DAG keeps 24,776 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 25,023 edges (0 negative), largest strong component 677; DAG keeps 21,886 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 20,452 edges (0 negative), largest strong component 621; DAG keeps 18,325 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 45,475 edges (20,452 negative), largest strong component 714.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 45,475 edges (20,452 negative), largest strong component 714.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 45,475 edges (20,452 negative), largest strong component 714.